

Huawei

Huawei is a leading Chinese manufacturer of smartphones and the largest manufacturer of telecommunication equipment in the world. HiSilicon is the branch of Huawei that manufactures SoCs for its smartphones and other devices. HiSilicon has purchased licenses from ARM for its cortex series of processors and Mali series of GPUs, which it uses on its SoCs.

Kirin 960

Announced early last year, Kirin 960 is the latest generation of high powered SoCs from Huawei which will power its latest flagship phone, the Mate 9. Based on the same big.LITTLE architecture that most of its competing SoCs are based on, the Kirin 960 is powered by a Cortex-A73 octa core CPU and a Mali-G71 graphics processor. Faster, smaller, more powerful and more efficient than its predecessor, the SoC has 4 high powered cortex-A73 cores clocked at 2.4 GHz to take care of intensive applications like gaming and audio/video editing, and 4 lower powered Cortex-A53 clocked at 1.8 GHz which consume less power and re used for various basic tasks.

The graphics performance of the chip is taken care of by the Mali-G71 MP8 GPU which is the first GPU in the new range of ARM's Bitfrost architecture based graphics processor. The 960's CPU is 15% more efficient than its predecessor's while the GPU featured on it is 20% faster.

Benchmarking tests reveal that Kirin 960 is more powerful than even the Apple A10 Fusion chip in multi core processing (Apple beat it at single core processing). Giving tough competition to other top of the line SoCs like Snapdragon 821 and Samsung Exynos 8890, the Kirin 960 packs a number of features like a faster UFS 2.1 storage chips that is capable of encoding and decoding HEVC videos at 4K resolution.

Primary CPU	4 x ARM Cortex A73 @2.4 GHz +4 x ARM Cortex A53 @ 1.8GHz
CPU architecture	ARM big.LITTLE
CPU structure	RISC
DSP type	Tensilica HiFi Audio DSP 4
GPU	ARM Mali-G71 MP8
Year Released	2016
Devices	Huawei Mate 9